

Instructions. The exam contains 3 questions and must be solved individually.

1. Define a sequential equilibrium. Please explain, intuitively, the difference between Sequential Equilibrium, Weakly Perfect Bayesian Equilibrium, and Extensive Form Perfect Equilibrium. (A single page suffices.)
2. For each $\epsilon > 0$, the following table describes a two-player normal-form game $\Gamma(\epsilon)$

		Player 2		
		e	f	g
Player 1	a	(0, 0)	(1, ϵ)	(ϵ , 1)
	b	(ϵ , 1)	(0, 0)	(1, ϵ)
	c	(1, ϵ)	(ϵ , 1)	(0, 0)

- a. Find all Nash equilibria of the game $\Gamma(\epsilon)$ for $\epsilon \neq 1$ but ϵ close to 1.
 - b. Assume $\epsilon = 1$. Find all Nash equilibria.
 - c. Assume $\epsilon = 1$. Identified a Trembling Hand Perfect equilibrium. Are there many?
3. Consider a standard principal-agent problem. Output y is the observable variable, effort $e \geq 0$ is the agent's "hidden" action. Suppose the principal offers a contract $t(y) = a + by$. If the agent accepts the contract t and output is y , the principal's net profit is $y - t(y)$. The principal is assumed to be risk neutral. If the agent accepts the contract t and chooses an effort level e , then the agent's expected utility is $U_A(e, t) = a + b\bar{y}(e) - \frac{1}{2}[b^2\sigma^2(e) + e^2]$ where $\bar{y}(e)$ and $\sigma^2(e)$ are the mean and variance of y given e .
If the agent rejects the contract, both agent and principal get a zero payoff.
Note that for the agent the effects of risk aversion and "effort aversion" are summarized by the last term of the agent's utility function.
 - a. Derive the principal's optimal choice of the two parameters a and b when output has the mean $\bar{y}(e) = e$ as a function of the agent's effort e , and the variance σ^2 of output is fixed independently of e .
 - b. Repeat the exercise for the case in which the principal observes directly the agent's effort e and therefore can make the parameters a and b depend on the actual effort e exerted by the agent.

$a(e) \quad b(e)$